

# **TITLE: Decentralized Inventory Management – Implement an inventory tracking system using blockchain.**

## **Introduction:**

In the dynamically changing environment of logistics and supply chain management, transparency and trust among players like suppliers, carriers, and retailers are paramount. Decentralized systems provide solutions to conventional centralized systems as they avoid fraud, delays, and anonymity.

This project shows a decentralized inventory and logistics management system based on blockchain technology, which ensures safe, transparent, and unalterable processing and tracking of orders between three central players — Shipper, Carrier, and Retailer.

## **Problem Statement:**

This project seeks to develop an open and secure inventory management system through the use of blockchain technology. Through the recording of inventory transactions on a distributed ledger, it provides data immutability and increases trust among stakeholders. The decentralized system minimizes dependency on central authorities, increasing efficiency and accountability in managing inventories. Finally, this project investigates the possibility of blockchain in transforming the way businesses manage and monitor their assets.

## **Blockchain Implementation Details:**

1. Smart Contracts
  - AccessControl.sol: Controls roles (Shipper, Carrier, Retailer) with limited permissions
  - BaseInventoryManagement.sol: Enables Shippers to manage inventory (items, quantity, pricing)
  - ComplexInventoryManagement.sol: Allows Retailers to order, and Carriers to report shipment and delivery status
2. Tools Used
  - Solidity for smart contracts
  - Truffle & Ganache for local development
  - React for frontend UI
  - MetaMask for Web3 wallet interaction
3. Deployment

Smart contracts are deployed via Truffle migrations, and the contract addresses/ABIs are incorporated into the frontend to enable real-time interaction through MetaMask.

## **SMART CONTRACTS IN DETAILED:**

1. **AccessControl.sol** contract handles permissions and roles in the system, using OpenZeppelin's AccessControl library. It specifies administrative functions and

ensures that only approved users (warehouse managers) can execute sensitive actions such as managing inventory.

2. **BaseInventoryManagement.sol** contract houses the main inventory logic. It enables registered managers to create warehouses, add items to them, and fetch item information. This contract applies access control to guarantee that only authorized users can manipulate inventory information.
3. **ComplexInventoryManagement.sol** contract is an addition to **baseinventorymanagement.sol** that adds a full supply chain layer, including roles like shippers, carriers, and retailers. It allows retailers to order, shippers to allocate carriers, and carriers to change the shipping status. Orders go through several states—from pending to shipped and eventually delivered—offering a transparent and decentralized method of tracking goods across the supply chain

## Required ScreenShots:

Compiling contracts:

```
PS C:\Users\MADHUMITHA V\CSE\SEM-6\BLOCKCHAIN\InventoryManagementDApp> truffle compile

Compiling your contracts...
=====
> Compiling .\contracts\Migrations.sol
> Compiling .\contracts\accesscontrol.sol
> Compiling .\contracts\baseinventorymanagement.sol
> Compiling .\contracts\complexinventorymanagement.sol
> Artifacts written to C:\Users\MADHUMITHA V\CSE\SEM-6\BLOCKCHAIN\InventoryManagementDApp\frontend\src\contracts
> Compiled successfully using:
   - solc: 0.8.0+commit.c7dfd78e.Emscripten.clang
PS C:\Users\MADHUMITHA V\CSE\SEM-6\BLOCKCHAIN\InventoryManagementDApp>
```

Deploying contracts

```
4_deployment_summary.js
=====

----- DEPLOYMENT SUMMARY -----
Contract Deployment Status:
- AccessControl: ✅ Deployed
- BaseInventoryManagement: ✅ Deployed
- ComplexInventoryManagement: ✅ Deployed

Contract Addresses:
- AccessControl: 0x3d75b0504Ef554898eC9443042bFf3062C5658df
- BaseInventoryManagement: 0x17089C8520d819817F240375cE7ca26De06DE081
- ComplexInventoryManagement: 0xEA29Ffef40C4b38816A21e7D5979cC707dFccB38
-----

> Total cost: 0 ETH

Summary
=====
> Total deployments: 3
> Final cost: 0.005982874768275003 ETH

PS C:\Users\MADHUMITHA V\CSE\SEM-6\BLOCKCHAIN\InventoryManagementDApp>
```

Launchpad 0 0 Connect Live Share BLACKBOX Chat Add Logs CyberCoder Improve Code Share Code Link Open V

DApp Website:

## Inventory Management DApp

### Add Warehouse Manager

### Register Warehouse

### Add Item to Warehouse

### Create Shipment

### Mark Shipment as Delivered

### Status

## FUTURE ENHANCEMENTS:

1. Adding more features to the frontend.
2. More debugging strategies.
3. Analytics dashboard for performance tracking.
4. IPFS for decentralized storage of product documents.